Optimizing Hospitality:
Data-Driven Insights into India's Hotel Economy

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Tools :

Python | Pandas | Matplotlib | Excel | Power BI | Chrome Web Scraping Extension

Cities Covered:
8 Tier-1 Indian Cities
(~150 hotels per city, merged dataset)

Contents:

1. Executive Summary
2. Project Objective
3. Data Overview
4. Methodology
 4.1 Data Collection
 4.2 Data Cleaning & Preprocessing
 4.3 Exploratory Data Analysis (EDA)
 4.4 Data Visualization & Dashboarding
5. Key Analysis & Insights
 5.1 Price Distribution
 5.2 Rating Distribution
 5.3 Price vs Rating Analysis
 5.4 City-Wise Hotel Performance
 5.5 Top 5 Hotels by City
6. Dashboard Summary
 6.1 Dashboard 1 – Price & Rating Analysis
 6.2 Dashboard 2 – Distribution & Top Hotels
7. Business Use Cases
8. Limitations
9. Business Recommendations
10. Conclusion

1. Executive Summary

The hotel industry is highly competitive, with pricing and customer ratings playing a critical role in booking decisions. This project focuses on analyzing hotel prices and ratings across major Tier-1 cities in India to identify pricing patterns, customer perception trends, and city-wise hotel performance.

Hotel data was collected using a Chrome web scraping extension, cleaned and processed using Excel and Python, and finally visualized using interactive Power BI dashboards. The analysis helps understand price distribution, rating behavior, and top-performing hotels across cities, enabling insights useful for travelers, hotel managers, and market analysts.

2. Project Objective

The primary objectives of this project are:

- Analyze hotel price distribution across major cities
- Understand customer rating patterns and trends
- Study the relationship between price and rating
- Identify top hotels within each city based on pricing and ratings
- Build interactive dashboards to support data-driven insights

3. Data Overview

Data Source

- Hotel data collected using a Chrome web scraping extension
- Data collected city-wise, then merged into a single dataset

Dataset Details

- ~150 hotels per city
- Total hotels ≈ 1200
- Key columns used:
    - Hotel Name
    - City
    - Price
    - Rating
    - Number of Reviews

Note: The dataset represents a sample of hotels, not all hotels in each city.

4. Methodology

4.1 Data Collection

- Hotel listings were scraped separately for each of the 8 Tier-1 cities
- Data was exported as CSV files and later merged

4.2 Data Cleaning & Preprocessing

Cleaning was done using Excel and Python, including:

- Removing duplicate hotel records
- Handling missing prices and ratings
- Converting price and rating columns to correct data types
- Standardizing city names and column formats

4.3 Exploratory Data Analysis (EDA)

EDA was performed using Python (Pandas & Matplotlib):

- Price distribution analysis
- Rating distribution analysis
- Price vs Rating relationship
- City-wise comparisons

These insights guided the selection of visuals for the Power BI dashboards.

4.4 Data Visualization & Dashboarding

- Cleaned dataset loaded into Power BI
- Interactive dashboards created with slicers for city filtering
- Focus on clarity, usability, and business relevance

5. Key Analysis & Insights

5.1 Price Distribution

- Most hotels fall within a mid-price range
- Premium hotels form a smaller percentage of total listings
- Price spread varies significantly across cities

5.2 Rating Distribution

- Majority of hotels are rated between 3.5 and 4.5
- Very low and very high ratings are relatively rare
- Rating concentration indicates generally positive customer experiences

5.3 Price vs Rating Analysis

- Higher price does not always guarantee higher rating
- Some mid-priced hotels achieve excellent ratings
- Indicates value-for-money opportunities for customers

5.4 City-Wise Hotel Performance

- Certain cities show higher average hotel prices
- Rating patterns differ by city
- Helps identify competitive and premium hotel markets

5.5 Top 5 Hotels by City

- Top hotels identified using:
  - Higher ratings
  - Competitive pricing
- Useful for recommendation and benchmarking analysis

6. Dashboard Summary

6.1 Dashboard 1 – Price & Rating Analysis

Includes:

- Price vs Rating (Scatter Plot)
- Average Price by City
- Rating distribution by city
- City filter slicer

Purpose:

- Understand how pricing relates to customer satisfaction

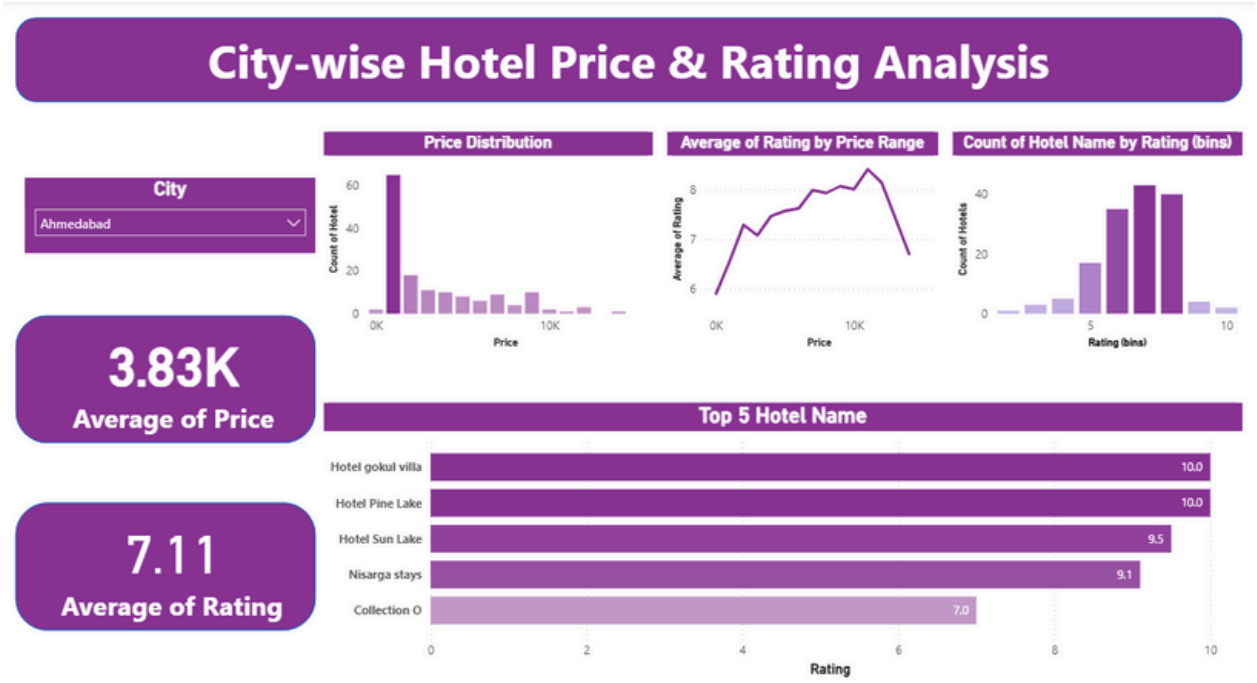
6.2 Dashboard 2 – Distribution & Top Hotels

Includes:

- Price Distribution (Histogram / Column Chart using binning)
- Rating Distribution (Column Chart)
- Top 5 Hotels by City
- City slicer for dynamic analysis

Purpose:

- Analyze overall pricing structure and Identify top-performing hotels per city



7. Business Use Cases

- Travel Platforms: Price benchmarking and hotel recommendations
- Hotel Owners: Competitive pricing and performance comparison
- Customers: Identifying value-for-money hotels
- Market Analysts: Understanding city-wise hotel market trends

8. Limitations

- Dataset contains a sample of hotels, not all hotels per city
- Prices may vary by date and availability
- Ratings are subject to customer bias and review volume

9. Business Recommendations

Based on the analysis of hotel pricing, ratings, and customer feedback across Tier-1 cities, the following actionable recommendations can be derived for key stakeholders:

9.1 Recommendations for Hotel Owners & Managers

1. Optimize Pricing Based on Market Positioning

- Hotels priced significantly higher than the city average but receiving average or low ratings should reassess pricing strategies.
- Mid-priced hotels with strong ratings represent an opportunity to increase prices marginally without impacting demand.

Action:

Use competitor pricing within the same city to align room rates with perceived value.

2. Improve Service Quality to Justify Premium Pricing

- Analysis shows that higher prices do not always correspond to higher ratings.
- Premium hotels with lower ratings should focus on service improvements rather than price increases.

Action:

Invest in customer experience improvements such as cleanliness, service responsiveness, and amenities to improve ratings.

3. Encourage Customer Reviews to Build Trust

- Hotels with higher review counts tend to show more stable ratings.
- Low-review hotels risk rating volatility.

Action:

Encourage guests to leave reviews post-stay through follow-up emails or incentives.

9.2 Recommendations for Online Travel Platforms (OTAs)

4. Promote Value-for-Money Hotels

- Several hotels with moderate prices show high ratings, indicating strong value-for-money offerings.

Action:

Highlight these hotels with badges such as “Best Value” or “Highly Rated & Affordable” to improve conversions.

5. Improve Sorting & Filtering Options

- Customers benefit when able to filter hotels by both rating and price range simultaneously.

Action:

Enable advanced filters like High Rating under ₹X or Top Rated Mid-range Hotels.

9.3 Recommendations for Travelers

6. Avoid Assuming Higher Price Equals Higher Quality

- The analysis indicates weak correlation between price and rating.

Action:

Travelers should focus on rating and review volume rather than price alone to maximize value.

7. Prefer Hotels with Balanced Ratings & Reviews

- Hotels with high ratings and sufficient review counts are more reliable choices.

Action:

Select hotels with ratings above 4 and a reasonable number of reviews for consistent experiences.

9.4 Recommendations for Market & Business Analysts

8. Use Sampling for Fair City Comparison

- Using a consistent sample size (150 hotels per city) allows fair comparative analysis.

Action:

For deeper insights, expand the dataset to include seasonal pricing and occupancy trends.

9. Enhance Future Analysis with Additional Metrics

- Adding variables like room type, location proximity, or amenities would strengthen analysis.

Action:

Incorporate additional dimensions for predictive pricing and demand forecasting models.

9.5 Strategic Insight Summary

- Price alone is not a strong indicator of hotel quality.
- Mid-range hotels frequently outperform premium hotels in customer satisfaction.
- Transparent pricing combined with quality service leads to higher ratings and customer trust.
- Data-driven pricing and review management can significantly improve hotel competitiveness.

10. Conclusion

This project demonstrates an end-to-end data analytics workflow, starting from web data collection to interactive dashboard creation. By combining Python-based EDA with Power BI visualizations, meaningful insights were generated around hotel pricing, ratings, and city-wise performance.

The dashboards provide an intuitive way to explore hotel market trends and support data-driven decision-making for multiple stakeholders.